BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-01-2026**

Course Name: Cybersphere and Security(CSS) **Page no:**

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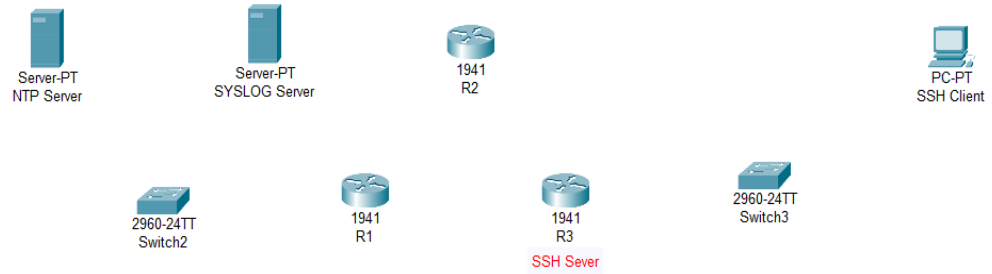
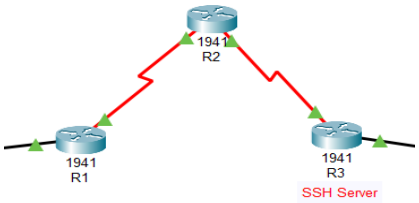
Q. Configure Routers: to support SSH connections.

1. Open the Cisco packet
2. Add the three routers of 1941 (R1, R2, R3) place it like triangle.
3. Click on each router (R1, R2, R3) → Go to the "Physical" tab → Turn off the router → Drag and drop the "HWIC-2T" module into an empty slot → Turn the router back on → Repeat for all routers.

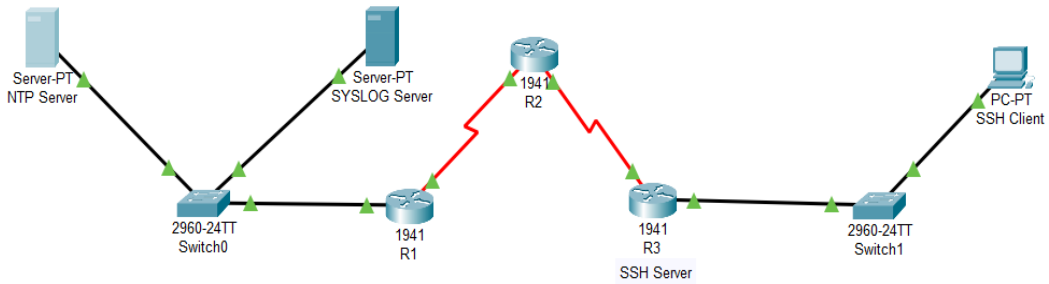
4. Take two 2960-24TT Switches → two servers → one PC and give it names as shown.



5. Connect the serial DTE wire to the router.



6. Connect the copper straight-through wire to devices as shown below.



7. Click on R1 and give addresses and on port status as shown, then close the tap.

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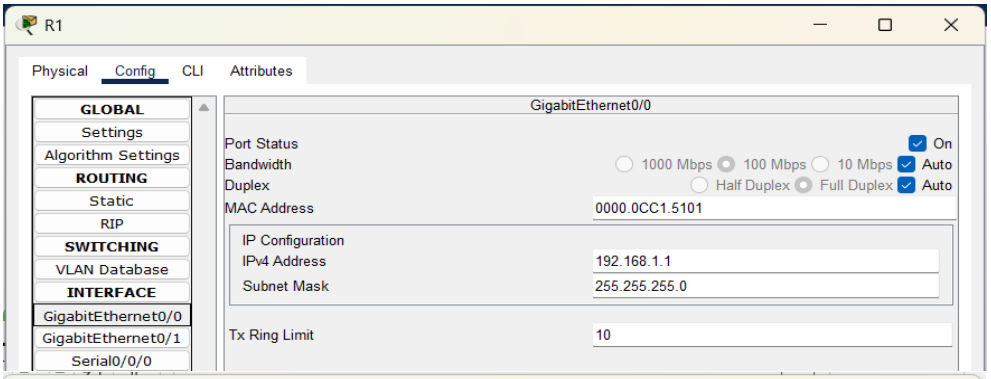
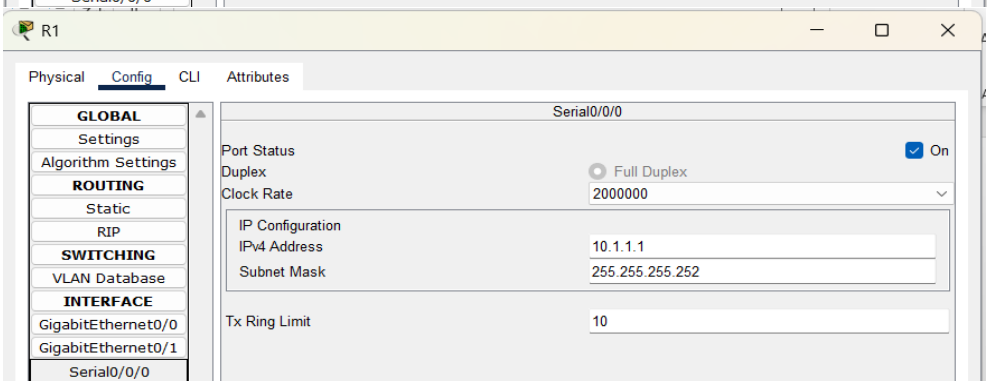
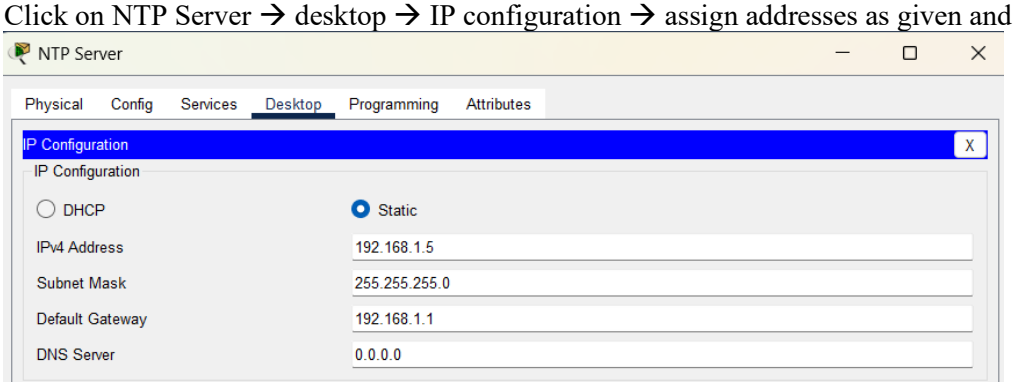
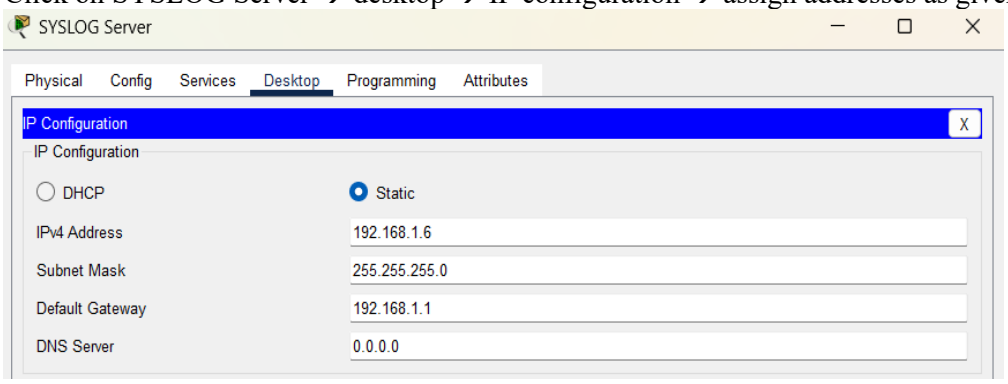




8. Click on NTP Server → desktop → IP configuration → assign addresses as given and close.



9. Click on SYSLOG Server → desktop → IP configuration → assign addresses as given and close.



10. Click on R2 and give addresses and on port status as shown, then close the tap.

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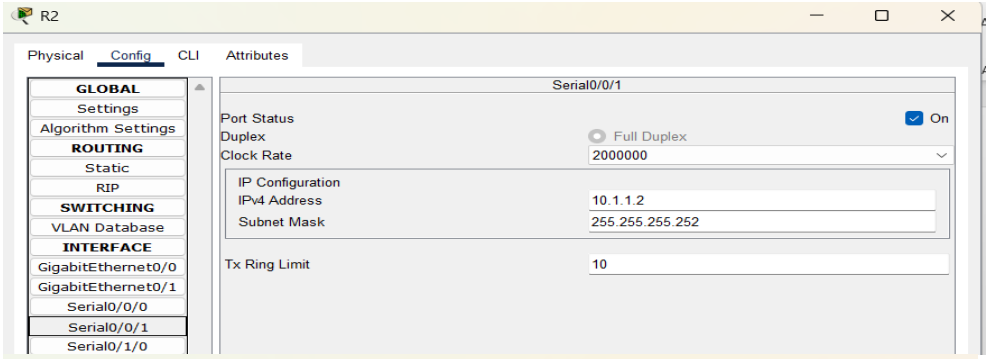
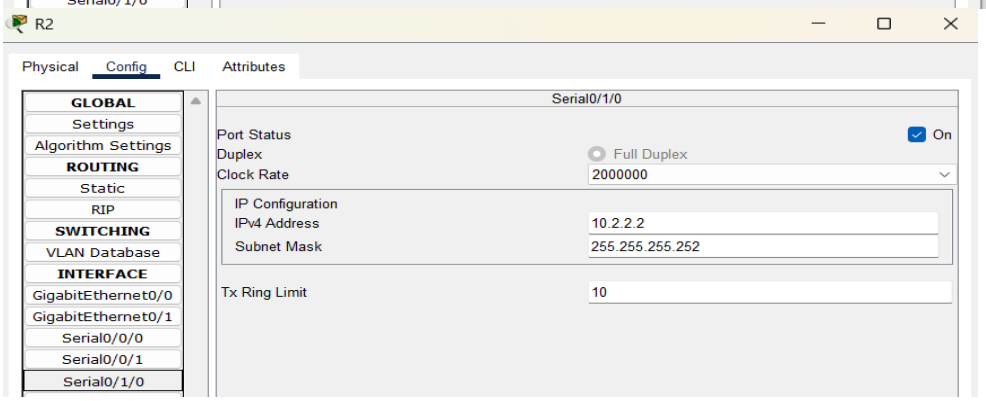
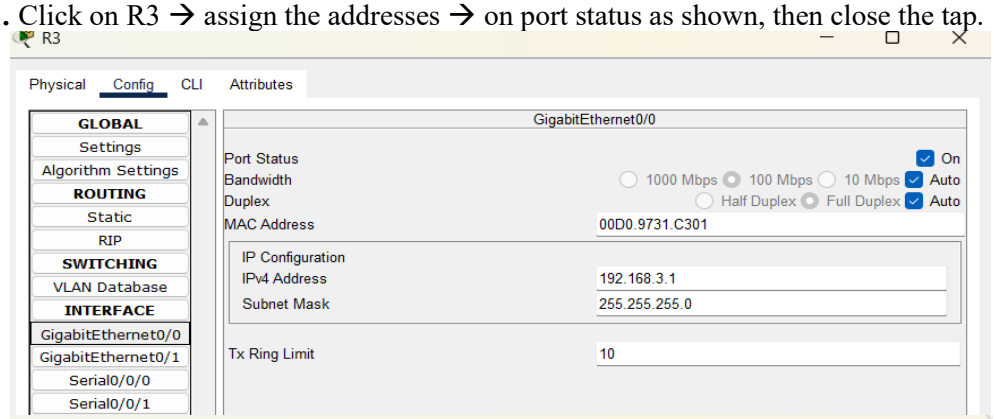
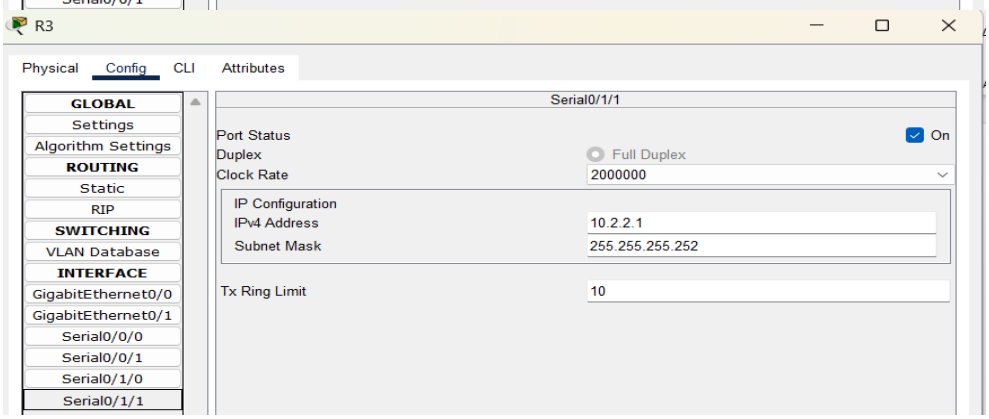




11. Click on R3 → assign the addresses → on port status as shown, then close the tap.





12. Click on SSH Client → desktop → IP configuration → assign addresses as given and close.

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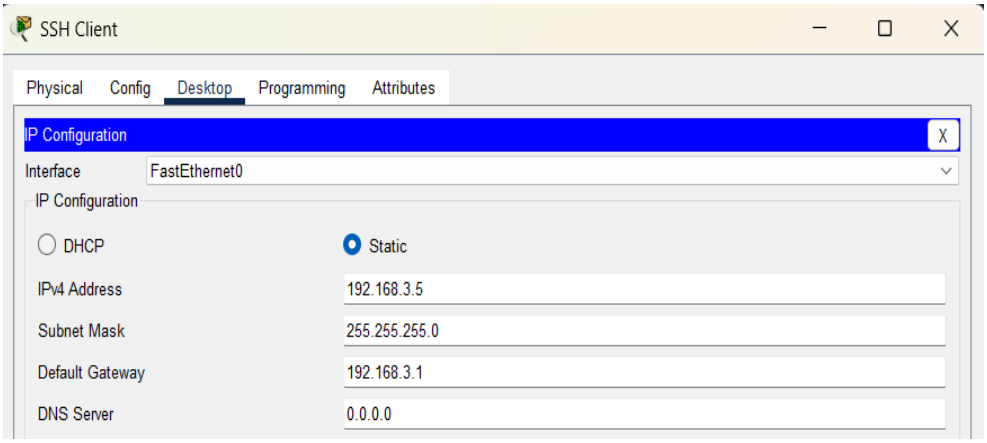
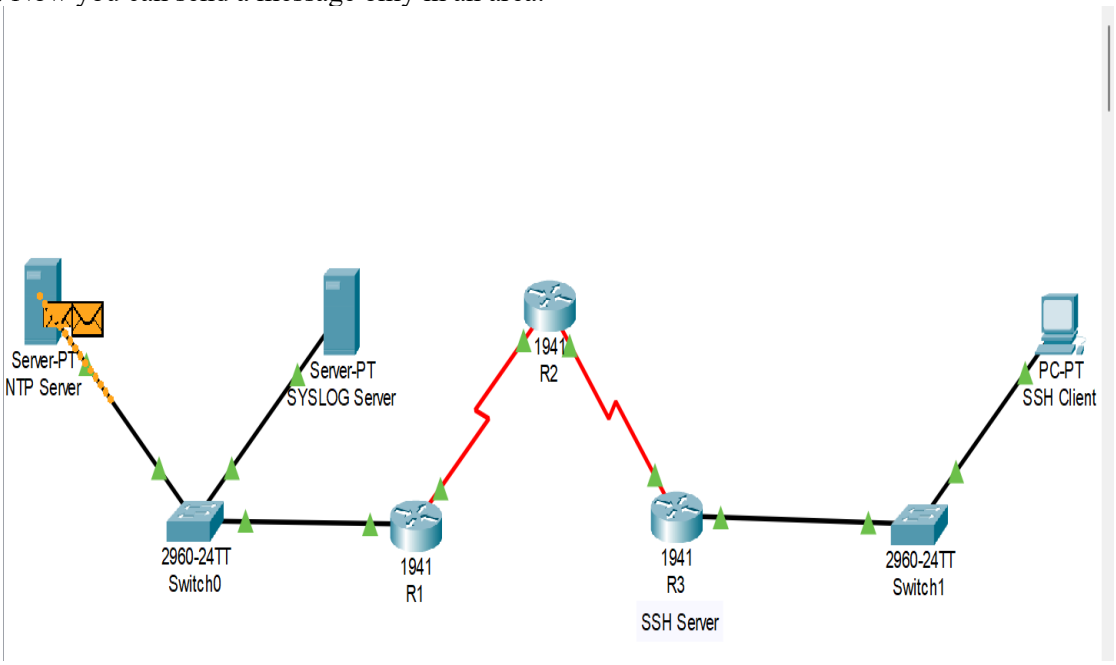


13. Now you can send a message only in an area.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	SYSL...	R1	ICMP		0.000	N	2	(edit)	(delete)
	Successful	R3	SSH Client	ICMP		0.000	N	3	(edit)	(delete)
	Successful	SSH ...	R3	ICMP		0.000	N	4	(edit)	(delete)

14. Click on R1 → CLI → paste these three lines in that, then close.

- router ospf 1
- network 192.168.1.0 0.0.0.255 area 0
- network 10.1.1.0 0.0.0.3 area 0

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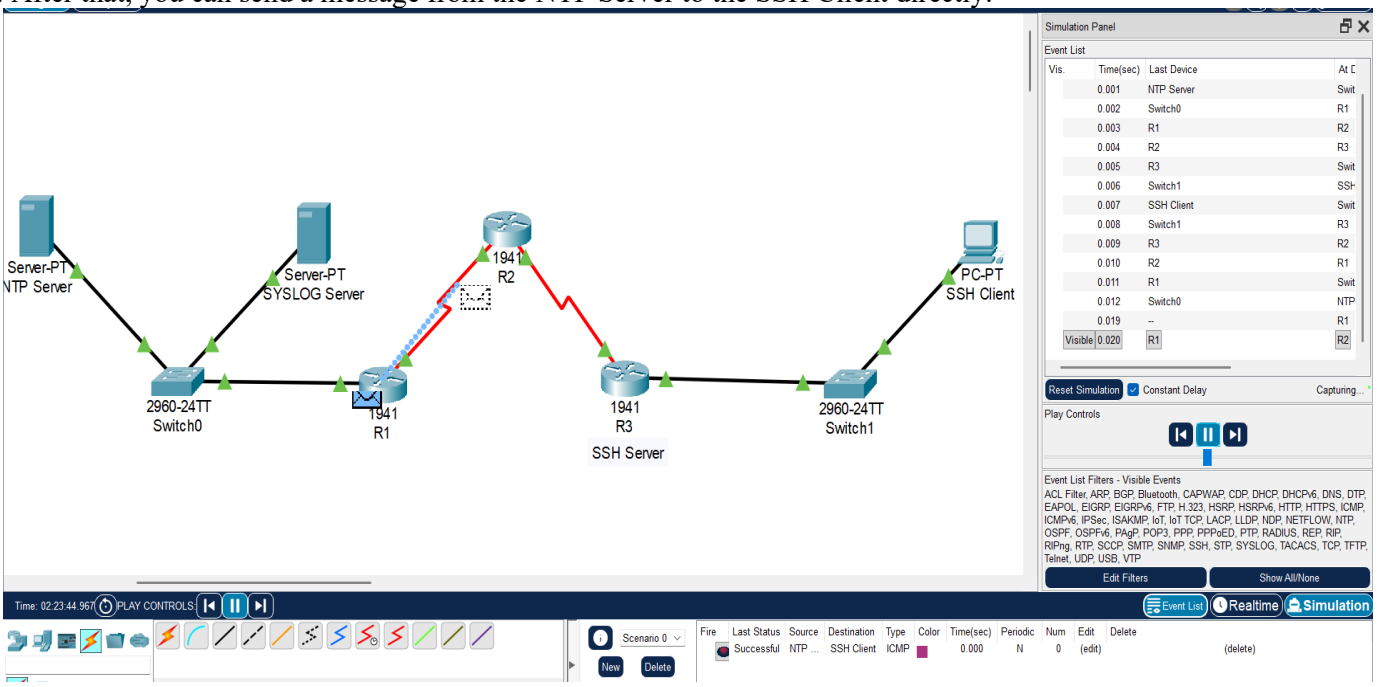
15. Click on R2 → CLI → paste these three lines in that, then close.
- router ospf 2
- network 10.1.1.0 0.0.0.3 area 0
- network 10.2.2.0 0.0.0.3 area 0

16. Click on R3 → CLI → paste these three lines in that, then close.
- router ospf 3
- network 192.168.3.0 0.0.0.255 area 0
- network 10.2.2.0 0.0.0.3 area 0

17. After that, you can send a message from the NTP Server to the SSH Client directly.



18. OSPF configuration :

R1:

```
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
```

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#ntwork 10.1.1.0 0.0.0.3 area 0
                      ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#
```

Password:

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```
R1>
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#area 0 authentication message-digest
R1(config-router)#exit
R1(config)#
```

Interface:

```
R1(config)#interface s0/1/0
R1(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R1(config-if)#exit
R1(config)#
```

Show ip ospf interface:

```
R1#show ip ospf interface
  Message digest authentication enabled
    Youngest key id is 1
R1#
```

R2:

```
Router(config-if)#exit
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
```

```
Router(config)#router ospf 2
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#
00:23:19: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
```

Password:

```
R2>en
Password:
00:48:41: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Dead timer expired

00:48:41: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Interface down or detached

Password:
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#area 0 authentication message-digest
R2(config-router)#exit
R2(config)#
```

Interface:

```
R2(config)#interface s0/1/1
R2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)#exit\
```

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R3:

```
Router(config-if)#exit
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
Router(config)#router ospf 3
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
```

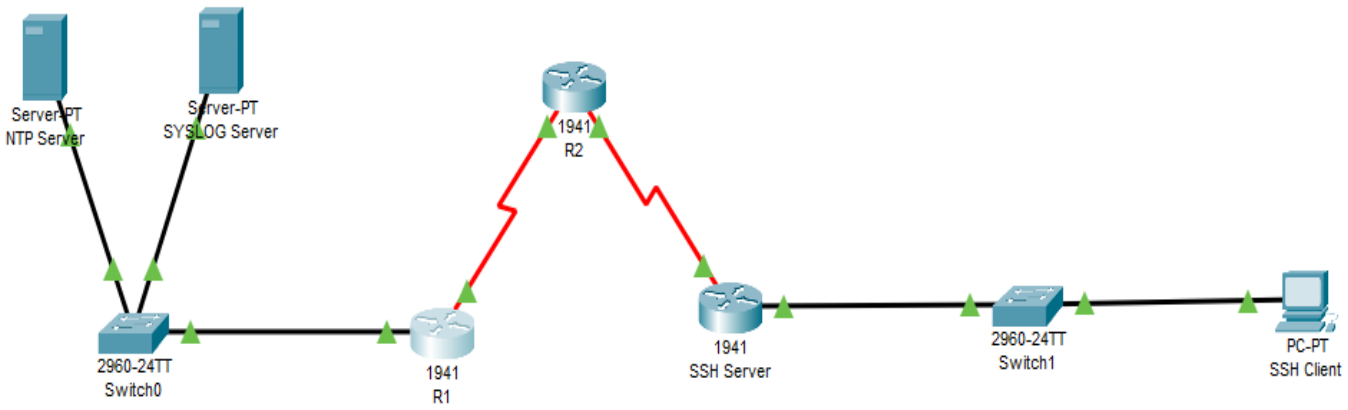
Password:

```
Router>en
Password:
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
Router(config-router)#exit
Router(config)#
```

Interface:

```
Router(config)#interface s0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
Router(config)#
```

Output:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Ec
	Successful	NTP S...	SSH Client	ICMP		0.000	N	0	(e
	Successful	SYSL...	SSH Client	ICMP		0.000	N	1	(e
	Successful	SSH ...	NTP Server	ICMP		0.000	N	2	(e
	Successful	SSH ...	SYSLOG ...	ICMP		0.000	N	3	(e

19. SSH Configuration:

- Click on R3 → CLI and paste the following commands.
  - conf t

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- ip domain-name ccnasecurity.com
- username SSHadmin privilege 15 secret ciscosshpa55
- line vty 0 4
- login local
- transport input ssh
- crypto key zeroize rsa
- crypto key generate rsa
- 1024
- ex
- show ip ssh

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip domain-name ccnasecurity.com
R3(config)#username SSHadmin privilege 15 secret ciscosshpa55
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#en
% Ambiguous command: "en"
R3(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.

R3(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.

R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#ex
*Dec 13 3:6:41.715: %SSH-5-ENABLED: SSH 2 has been enabled
R3#
*Dec 13, 03:06:46.066: SYS-5-CONFIG_I: Configured from console by console
R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
```

- Now write the following commands in the same R3.
  - conf t
  - ip ssh time-out 90
  - ip ssh authentication-retries 2
  - ip ssh version 2
  - ex
  - show ip ssh

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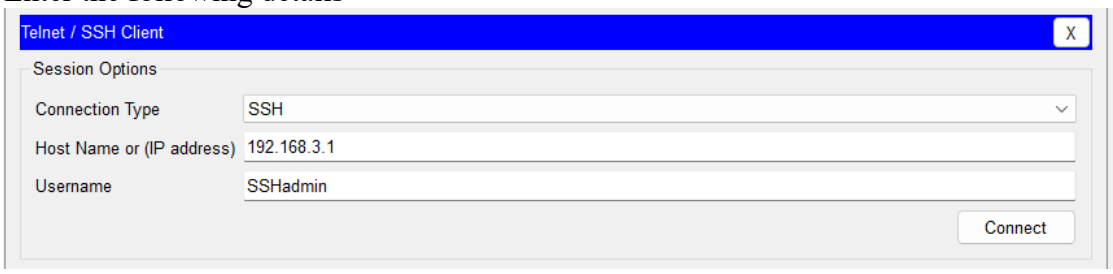
```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#ip ssh version 2
R3(config)#ex
R3#show ip ssh
*Dec 13, 03:07:10.077: SYS-5-CONFIG_I: Configured from console by console
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R3#
```

20. Verification

- Click on SSH Client PC → Desktop → Telnet/SSH Client
- Enter the following details



- Click on connect

- Enter password as **ciscosshpa55** and hit enter

- Type command **show ip ssh**

```
R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R3#
```